Sahaj Bhadja

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Summary

Quantitative **Computer Science** candidate (MSCS, USC) with 2+ years analyzing **software systems**, building **data models** and analytics pipelines, and producing research-grade **technical reports**, with deep **software-sector** knowledge and fintech, capital-markets-adjacent platform experience.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built and maintained **data models and relational databases** for 500+ users and 10K+ structured records, designing **analytics dashboards** and reducing query latency from 1.2s to 350ms for faster reporting.
- Developed **ETL data pipelines** consolidating sources into normalized datasets with validation and reconciliation checks across 5K+ monthly jobs, improving data accuracy and reliability for downstream analysis and reporting.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered an **AI analytics platform** processing 10K+ daily requests and built **cost-tracking models** quantifying per-operation spend, enabling data-driven decisions across high-volume software product workflows.
- Translated **product and stakeholder requirements** into shipped features for 150+ users, communicating technical tradeoffs clearly and meeting time-sensitive deadlines in a fast-paced startup environment.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Conducted **quantitative research** over 100K+ records, building evaluation models and benchmarking pipelines that produced reproducible, high-confidence findings on model-driven decision performance and generalization.
- Authored **detailed technical research reports** and comparative analyses, translating complex quantitative findings into clear, actionable guidance that improved cross-team alignment and decision-making.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Led **ETL pipeline redesign** with AWS Glue and S3, consolidating multi-source data into normalized formats for analytics and reporting, improving pipeline efficiency **68%** and reducing manual data preparation.
- Optimized large-scale **data integrations** in an Agile environment with Microsoft/GitHub stakeholders, improving response times **30%** while maintaining detailed documentation and quality across coordinated deliverables.

Cyber Security and Data Mining: Detecting Malicious URL **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*
- Analyzed a **650K+ record dataset** through statistical **feature engineering** across 21 behavioral indicators, building classification models that reached 96%+ accuracy and surfaced actionable insights into emerging risk patterns.
- Generated **research findings and data-driven reports** comparing multiple analytical models, translating quantitative results into clear written conclusions on detection trends and risk exposure.

Skills

Quantitative & Research: Quantitative Analysis, Statistical Analysis, Data Modeling, Feature Engineering, Benchmarking, Research & Reporting, Technical Writing
Tools: Microsoft Excel, Microsoft Office, Tableau, Splunk, Jupyter, Git
Data & Databases: SQL, PostgreSQL, MySQL, ETL Pipelines, Databricks, Spark, Data Warehousing
Software Sector: SaaS, Cloud (AWS, Azure), APIs, Microservices, Fintech, Distributed Systems
Programming: Python, Java, JavaScript/TypeScript, Bash
Markets & Domain: Capital Markets (familiarity), Payment Systems, Financial Platforms, Risk Analysis

Projects

FinNexus Live | *Fintech Platform, Real-Time Data, Plaid*
- Built a **fintech analytics platform** processing 1M+ transactions for 50K+ users with real-time data feeds, secure access, and advanced filtering, surfacing transaction insights across high-volume financial workflows.

Big-Brain Live | *Document Research, Semantic Search, Reporting*
- Built an **AI document-research platform** over 50K+ documents enabling semantic search, summarization, and Q&A, reducing document-analysis time **60%** to accelerate report generation and information sourcing.

Chat2SQL Code | *SQL, Databases, Data Analysis*
- Built a **data workspace** supporting 10+ databases with schema-aware querying and reporting, accelerating data sourcing, exploration, and structured analysis from a single unified interface.